

Ⓐ For an LTI system the impulse response $\rightarrow h(t) = \frac{\sin(\omega_0 t)}{t}$. Find the output $y(t)$ for an input $x(t) = \frac{\sin(2\omega_0 t)}{t}$

We should know that

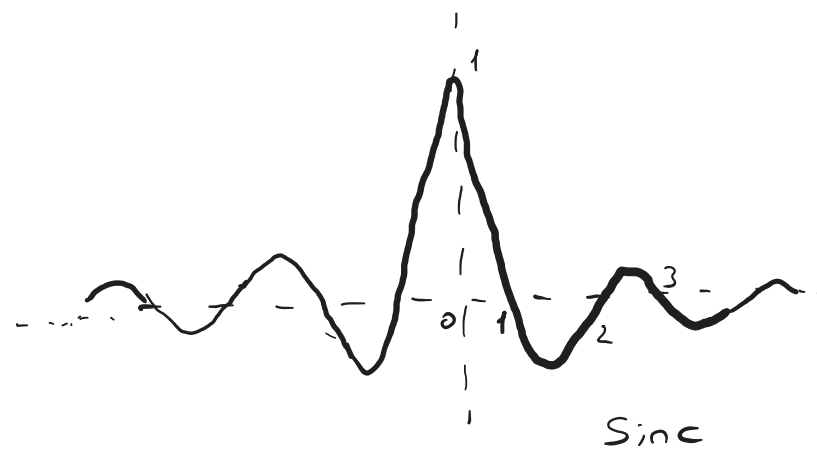
• $\text{rect}(t) = \begin{cases} 1, & |t| < \frac{1}{2} \\ 0, & |t| > \frac{1}{2} \end{cases}$ for the pulse,

$\text{sinc } f = \frac{\sin \pi f}{\pi f}$ for its spectrum

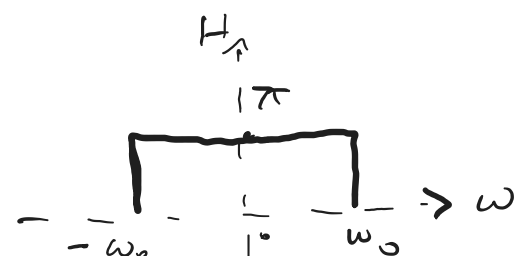
$\mathcal{F}\left\{\frac{\sin(\omega_0 t)}{t}\right\} = \begin{cases} 1, & |\omega| \leq \omega_0 \\ 0, & |\omega| > \omega_0 \end{cases}$

$H(j\omega) = \mathcal{F}[h(t)] = \mathcal{F}\left\{\frac{\sin(\omega_0 t)}{t}\right\}$

$= \begin{cases} \pi, & |\omega| \leq \omega_0 \\ 0, & |\omega| > \omega_0 \end{cases}$

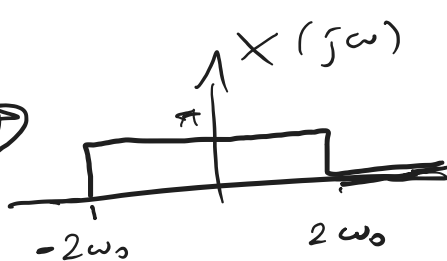


$\text{sinc } x = \frac{\sin x}{x}$



$X(j\omega) = \mathcal{F}[x(t)] = \mathcal{F}\left\{\frac{\sin(2\omega_0 t)}{t}\right\}$

$= \begin{cases} \pi, & |\omega| \leq 2\omega_0 \\ 0, & |\omega| > 2\omega_0 \end{cases}$



Remember!

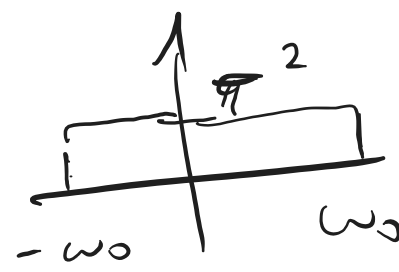
Impulse resp. input and output relationship?

Convolution

$y(t) = x(t) * h(t)$

$Y(j\omega) = \mathcal{F}[y(t)] = X(j\omega) H(j\omega)$

$= \begin{cases} \pi^2, & |\omega| \leq \omega_0 \\ 0, & |\omega| > \omega_0 \end{cases}$



$Y(j\omega) \xrightarrow[\text{FT}]{\text{inverse}} y(t) = \frac{1}{2\pi} \int_{-\infty}^{\infty} Y(\omega) e^{j\omega t} d\omega$

$= \frac{1}{2\pi} \int_{-\omega_0}^{\omega_0} \pi^2 e^{j\omega t} d\omega$

$= \frac{\pi^2}{2\pi} \int_{-\omega_0}^{\omega_0} e^{j\omega t} d\omega$

$= \frac{\pi}{2} \left(\frac{e^{j\omega_0 t} - e^{-j\omega_0 t}}{j\omega} \right)$

$= \frac{\pi}{2} \cancel{2} \frac{\sin(\omega_0 t)}{t}$

$= \frac{\pi \sin(\omega_0 t)}{t}$

↗ Euler's identity

Signals and Systems

Q1 Find the DTFT of $x(n) = \frac{(n+m-1)!}{n!(m-1)!} a^n u(n)$, $|a| < 1$

Solution

let $x_1(n) = a^n u(n)$

$$X_1(e^{j\omega}) = \sum_{n=0}^{\infty} (a^n) e^{-j\omega n} = \sum_{n=0}^{\infty} (ae^{-j\omega})^n = \frac{1}{1 - ae^{-j\omega}}$$

$$x(n) = \frac{(n+m-1)!}{n!(m-1)!} a^n u(n)$$

for $m=1 \rightarrow x(n) = a^n u(n)$

for $m=2 \rightarrow x(n) = \frac{(n+1)!}{n!1!} = \frac{(n+1)n!}{n!} = n+1 \rightarrow (n+1) \cdot a^n u(n) \xrightarrow{FT}$

for $m=3 \Rightarrow x(n) = \frac{(n+2)!}{n!2!} = \frac{(n+2)(n+1)n!}{n!2!} = \frac{(n+2)(n+1)}{2}$

differentiation property

$$na^n u(n) \rightarrow j \frac{dX_1(e^{j\omega})}{d\omega} = j \frac{d}{d\omega} \left(\frac{1}{1 - ae^{-j\omega}} \right) = \frac{ae^{-j\omega}}{(1 - ae^{-j\omega})^2}$$

$$(n+1)a^n u(n)$$

$$= na^n u(n) + a^n u(n)$$

$$= \frac{ae^{-j\omega}}{(1 - ae^{-j\omega})^2} + \frac{1}{1 - ae^{-j\omega}}$$

$$= \frac{1}{(1 - ae^{-j\omega})^2}$$

$$m=2$$

$$\frac{n^2 + 3n + 2}{2} \Rightarrow \left(\frac{n^2 a^n u(n) + 3n a^n u(n) + 2 a^n u(n)}{2} \right)$$

$$\frac{1}{2} \left(\underbrace{n \cdot \frac{d}{dn} a^n u(n)}_{\text{derivative}} + 3 \cdot n a^n u(n) + 2 a^n u(n) \right)$$

$$\frac{1}{2} \left(\frac{ae^{-j\omega}(1 + ae^{-j\omega})}{(1 - ae^{-j\omega})^3} + \frac{3ae^{-j\omega}}{(1 - ae^{-j\omega})^2} + \frac{2}{(1 - ae^{-j\omega})} \right)$$

$$\frac{1}{2} \left(\frac{2}{(1 - ae^{-j\omega})^3} \right) = \frac{1}{(1 - ae^{-j\omega})^3} \rightarrow m=3$$

$\Rightarrow$ therefore

$$X(e^{j\omega}) = \frac{1}{(1 - ae^{-j\omega})^k}, k \in \mathbb{Z}$$

Q2 Prove $\sum_{n=-\infty}^{\infty} [x(n)]^2 = \frac{1}{2\pi} \int_{-\pi}^{\pi} |X(e^{j\omega})|^2 d\omega$

$\sum_{n=-\infty}^{\infty} x(n) x^*(n) \Rightarrow x(n) = \frac{1}{2\pi} \int_{-\pi}^{\pi} X(e^{j\omega}) e^{j\omega n} d\omega$ (Inverse FT)

$x^*(n) = \frac{1}{2\pi} \int_{-\pi}^{\pi} X^*(e^{j\omega}) e^{-j\omega n} d\omega$

Therefore

$\sum_{n=-\infty}^{\infty} x(n) \frac{1}{2\pi} \int_{-\pi}^{\pi} X^*(e^{j\omega}) e^{-j\omega n} d\omega$

$\rightarrow \frac{1}{2\pi} \int_{-\pi}^{\pi} X^*(e^{j\omega}) \underbrace{\sum_{n=-\infty}^{\infty} x(n) e^{-j\omega n} d\omega}_{}$

$= \frac{1}{2\pi} \int_{-\pi}^{\pi} X^*(e^{j\omega}) X(e^{j\omega}) d\omega$

$= \frac{1}{2\pi} \int_{-\pi}^{\pi} |X(e^{j\omega})|^2 d\omega$

✓

Q

$$x(t) \xrightarrow{\text{FT}} X(j\omega) \quad \text{then prove} \quad \int_{-\infty}^t x(\tau) d\tau \xrightarrow{\text{FT}} \frac{1}{j\omega} X(j\omega)$$

for $y = \int_{-\infty}^t x(\tau) d\tau \rightarrow$ differentiate both sides

$$\frac{d}{dt} y(t) = x(t)$$

$$j\omega \cdot Y(j\omega) = X(j\omega) \quad \checkmark$$

easy example but
it can indicate the main
strategy for proofs:

Start $\rightarrow$ Finish

Find
the
way

$\rightarrow$ manipulate
input
to get
similar something
to output //

② $x(t) \xrightarrow{FT} X(j\omega) \Rightarrow X(j\omega) = \begin{cases} 2, & |\omega| \leq 1 \\ 1, & 1 < |\omega| \leq 2 \\ 0, & \text{otherwise} \end{cases}$

a) determine the energy of signal $x(t)$

b) Find $x(t)$

Solution

a)

$$E = \int_{-\infty}^{\infty} |x(t)|^2 dt = \frac{1}{2\pi} \int_{-\infty}^{\infty} |X(j\omega)|^2 d\omega$$

$$= \frac{1}{2\pi} \left[\int_{-\infty}^{-2} |X(j\omega)|^2 d\omega + \int_{-2}^{-1} |X(j\omega)|^2 d\omega + \int_{-1}^1 |X(j\omega)|^2 d\omega + \int_1^2 |X(j\omega)|^2 d\omega + \int_2^{\infty} |X(j\omega)|^2 d\omega \right]$$

$$= \frac{1}{2\pi} \left[0 + \int_{-2}^{-1} 1 d\omega + \int_{-1}^1 4 d\omega + \int_1^2 1 d\omega + 0 \right]$$

$$= \frac{1}{2\pi} \left[(-1 - (-2)) + (4(1) - (-4)) + (2 - 1) \right]$$

$$= \frac{1}{2\pi} + \frac{8}{2\pi} + \frac{1}{2\pi} = \frac{5}{\pi}$$

b) Let's analyze the function first

$$X(j\omega) = \begin{cases} 2, & |\omega| \leq 1 \\ 1, & 1 < |\omega| \leq 2 \\ 0, & \text{otherwise} \end{cases}$$

$$X_1 = \begin{cases} 1, & |\omega| \leq 1 \\ 0, & |\omega| > 1 \end{cases}$$

$$+ X_2 = \begin{cases} 1, & 1 < |\omega| \leq 2 \\ 0, & |\omega| > 2 \end{cases}$$

Let's analyze

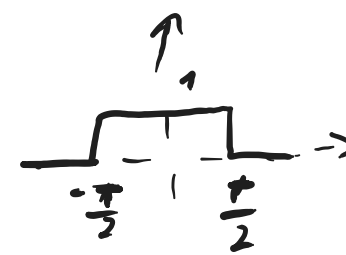
We have two parts

Both are rectangular pulse functions

Fourier transform of it, you don't need to memorize!

Let's define a rect. function like

$$x(t) = \begin{cases} 1, & |t| \leq \frac{T}{2} \\ 0, & \text{otherwise} \end{cases}$$



FT formula:

$$X(j\omega) = \int_{-\infty}^{\infty} x(t) e^{-j\omega t} dt = \int_{-T/2}^{T/2} x(t) e^{-j\omega t} dt$$

$$= \left(\frac{e^{-j\omega t}}{-j\omega} \right) \Big|_{-T/2}^{T/2} = -\frac{1}{j\omega} \left[e^{-j\omega \frac{T}{2}} - e^{+j\omega \frac{T}{2}} \right]$$

(Euler's identity)

$$e^{-j\theta} - e^{+j\theta} = -2j \sin \theta$$

Therefore we have

$$X(\omega) = -\frac{1}{j\omega} (-2j) \sin\left(\frac{\omega T}{2}\right)$$

$$= \frac{2}{\omega} \sin\left(\frac{\omega T}{2}\right) \rightarrow \frac{\sin\left(\frac{\omega T}{2}\right)}{\frac{\omega}{2}}$$

$$\boxed{\text{Sinc}(x) = \frac{\sin(\pi x)}{\pi x}}$$

(see the next question)

Therefore, finally we get

$$\frac{\sin(t)}{\pi t} + \frac{\sin(2t)}{\pi t}$$